

Computational Physics 2: Exercise 7

Giovanni Settanni, Marialore Sulpizi

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1 Molecular dynamics of the Lennard-Jones fluid

- Complete the python code in the file `md_vel2verlet_opt2_ex.py` to perform molecular dynamics simulations of the LJ-fluid.
- Run the code and check for conservation of total energy and momentum by plotting the time series of the two quantities.
- From the time series of temperature and potential energy, make hypothesis about what is happening in the simulation.
- Download and install VMD from the website of the University of Illinois (<https://www.ks.uiuc.edu/Research/vmd/>) and load the trajectory saved in the ".xyz" file for visualization. Follow the instruction in the code comments to enforce the wrapping of the coordinates in the periodic box and zoom to it with the mouse wheel. To observe more clearly the position of the particles you may need to change representation in the Graphic Representation to (VdW). You can move along the trajectory by moving the slider on the Main window.
- Make observations about what you see in the trajectory, especially regarding the local density of particles, and relate it to the behavior of the potential energy.
- Rename the trajectory (the program will otherwise append the new trajectory to the old one) and output file and run the code again after changing the initial temperature to a value above 1. Then repeat the previous steps and explain the observed changes in behavior.

2 Radial distribution function

- Complete the missing parts of the code in the file `gofrnjit_ex.py`.
- Run on the two trajectories obtained at the previous step.
- Plot the two radial distribution functions and explain the observed differences.